Shreya Ponnamma

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Professional Summary

M.Sc. Computer Science student at the University of Freiburg with more than four years of software engineering experience across industry and research environments. Experienced in backend development, machine learning pipelines, deep learning systems, AI-driven data engineering, and large-scale data processing. Currently working on transformer-based biomedical machine translation and semantic search systems.

Technical Skills

Programming Languages: Python, Java, SQL, C#, JavaScript, TypeScript

AI & Machine Learning: PyTorch, Deep Learning, Transformers, LLMs, Self-Supervised Learning, Domain Adaptation

Backend & Data Engineering: FastAPI, REST APIs, PostgreSQL, Elasticsearch, ETL Pipelines

Tools & Platforms: Docker, Git, Linux

Professional Experience

University Hospital Freiburg (TalaMed Project) Sep 2025 – Present
Working Student (AI, Backend & Data Engineering)

- Developed Python-based ETL pipelines for biomedical document processing and semantic search.
- Built backend services using FastAPI, REST APIs, PostgreSQL, Elasticsearch, and Docker.
- Designed data ingestion workflows for large-scale text collections.
- Improved reliability and maintainability of processing pipelines through testing and debugging.
- Collaborated with clinicians and researchers to support AI-driven medical search applications.

Fraunhofer EMI Jul 2025 – Present
Working Student (Data Engineering & Machine Learning)

- Developed Python workflows for scientific data preprocessing and validation.
- Automated extraction and transformation of heterogeneous research data.
- Improved reproducibility and maintainability of analysis pipelines.

Fraunhofer ISE Dec 2023 – May 2025
Research Assistant – Backend Systems and Data Engineering

- Developed backend components and database-driven applications.
- Worked with C#, .NET, and SQL Server for backend development tasks.
- Implemented data processing workflows and maintained software modules.
- Refactored existing modules to improve maintainability and performance.
- Supported testing, debugging, and documentation activities.

Finflux (M2P) Jul 2022 – Sep 2023
Software Engineer

- Developed backend services and REST APIs for financial applications.
- Implemented business logic and supported feature delivery in Agile teams.
- Participated in debugging and maintenance activities for production systems.

- Received Best Performer Award for Aadhaar-based e-KYC automation.

Aurigo Software Technologies Aug 2021 – Jul 2022
Software Engineer I

- Developed enterprise applications using C#, ASP.NET MVC, and SQL Server.
- Implemented new features and enhanced existing application modules.
- Performed debugging and production issue resolution.
- Participated in testing and code review activities.
- Received SPOT Award for automating SQL generation workflows.

Education

University of Freiburg, Germany Oct 2023 – Mar 2027 (expected)
M.Sc. Computer Science

Master's Thesis: Transformer-based Biomedical Machine Translation with Domain Adaptation

- Developing transformer-based models for German-English biomedical machine translation.
- Investigating parameter-efficient fine-tuning methods including LoRA and adapters.
- Evaluating domain adaptation strategies for heterogeneous clinical datasets.

Vidyavardhaka College of Engineering, India Aug 2017 – Aug 2021
B.E. Computer Science and Engineering

Projects

Biomedical Search Engine ETL Framework

- Developed ETL pipelines for large-scale ingestion and processing of biomedical literature and clinical text collections.
- Implemented structured extraction, normalization, and indexing workflows using Python.
- Integrated PostgreSQL and Elasticsearch to support scalable semantic search and information retrieval.
- Containerized services with Docker to ensure reproducibility and simplified deployment.

Microglial Image Analysis Pipeline

- Developed deep learning pipelines for the processing and analysis of more than 279,000 immunofluorescence microscopy images.
- Implemented image preprocessing, denoising, and feature extraction workflows using PyTorch and OpenCV.
- Investigated self-supervised learning approaches for improving image quality in low signal-to-noise microscopy datasets.
- Established standardized processing pipelines to ensure reproducibility across experiments.
- Contributed computational analysis results to interdisciplinary neuroscience research published in *iScience*.

Selected Publications

Ponnamma S., et al.
Re-engineered TalaMed Medical Search Engine: Clinical Data Integration and ETL Pipelines
ICIMTH 2026 (Accepted)

Jariwala V.D., **Ponnamma S.**, et al.
Spatiotemporal Analysis of Microglial Morphometrics Following Brain Microinjury
iScience